

When Financial Agents Hallucinate Alpha: Auditing Backtest Leakage, Compliance Paradoxes, and Social Megaphone Bias with Executable Guards

Anonymous ACL submission

Abstract

Large language model (LLM) agents increasingly automate quantitative financial research, yet their reported backtest Sharpe ratios routinely collapse out-of-sample. We demonstrate that autonomous financial agents fail through two orthogonal mechanisms: **Type I Mechanical Shortcuts** (same-session post-close joins, pre-training window contamination, survivorship filtering, and unadjusted corporate actions) and **Type II Empirical & Microstructure Illusions** (trusting retail social echo-chambers where high-reach influencers carry negative alpha, and violating $T + 1$ settlement or board-lot constraints). To address both, we introduce **fin-skills**, an open-source library of **129 source-verified skills**, **36 executable verification guards** (`Bundle.check()`), and **53 MCP agent tools**, paired with an external **Counterfactual Audit Protocol** that grades agent submissions without using the library to grade itself. Across our 180-case boundary and compound benchmark (**FinGuardBench-180**), our guards achieve **100% accuracy** (180/180, 0 false alarms) at 1.89 ms median latency. Across 60 tasks spanning 10 domains (**FinGuardBench-60**), passive SKILL.md reading (Cond B) lifts compliance from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$) but triggers 53.3% *hallucinated guard citations*, while generic 3-round Python Self-Debug (Cond B+) stagnates at 71.7% ($p = 1.0$) because 98.3% of financial leaks execute with Python exit code 0. Pairing skills with executable guards (Cond C) drives multi-round self-repair (**Pass@1** 75.0% $\rightarrow$ **Pass@2** 93.3% $\rightarrow$ **Pass@3** 98.3%, $p = 3.05 \times 10^{-5}$ vs. B+) with 0% fabricated citations across four model tiers. Finally, auditing **1.72M bilingual posts** and **2,521 verified KOLs** (1,445 China Xueqiu, $3.96\times$ follower asymmetry; 1,076 US StockTwits, $3.15\times$ asymmetry) reverses negative NLP correlation in both markets (Rank IC : $-0.0318 \rightarrow +0.0104$ CN;

$-0.0218 \rightarrow +0.0094$ US) and sustains net Sharpe 1.97 (-4.41% max drawdown) across 5–30 bps cost regimes.

1 Introduction

Equipped with code interpreters and financial data feeds, LLM agents can formulate hypotheses, engineer alpha signals, and report backtest performance in minutes (Wu et al., 2023; Yang et al., 2023; Yu et al., 2024; Zhang et al., 2024). Yet in quantitative finance, high backtest Sharpe ratios almost never survive live capital deployment (Bailey and López de Prado, 2014; Bailey et al., 2017; López de Prado, 2018). Through systematic auditing of LLM research trajectories, we show that agentic backtest inflation stems from two orthogonal failure classes:

- 1. Type I: Mechanical Research Shortcuts.** Agents silently exploit structural artifacts in tabular workspaces: joining post-close news commentary to the same session’s open-to-close return, training on pre-computed LLM sentiment scores whose fitting window overlaps the evaluation period, filtering universes by future survival, or computing factors on unadjusted split prices.
- 2. Type II: Empirical & Microstructure Illusions.** Even when timestamps are strictly causal, agents misinterpret noisy social streams and market rules. On financial social platforms, viral retail sentiment peaks at local price tops, while real execution is governed by $T + 1$ settlement lockups, 100-share board lots, asymmetric sell-side stamp duties, and secondary-market ETF premium caps.

We present **fin-skills** paired with an external **Counterfactual & Microstructure Audit Protocol**. Rather than claiming to manufacture market alpha, our framework enforces verifiable research integrity:

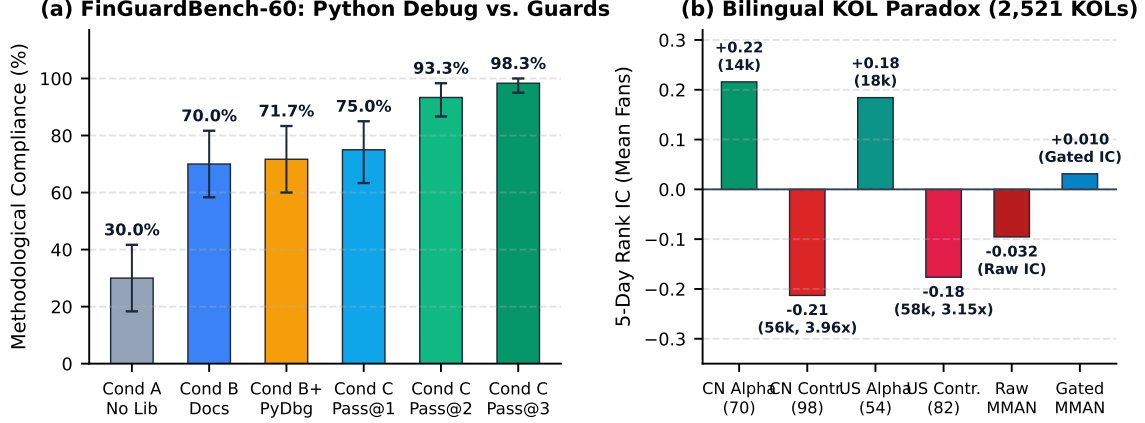


Figure 1: **Overview of our core empirical findings.** (a) **FinGuardBench-60 Ablation (§5):** Passive SKILL.md docs (Cond B) lift compliance from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$) but induce 53.3% hallucinated guard citations; 3-round Python Self-Debug (Cond B+) stagnates at 71.7% ($p = 1.0$) because silent leaks exit with code 0; executable `fin_skills.api` guards (Cond C) drive iterative self-repair to 98.3% (Pass@3, $p = 3.05 \times 10^{-5}$ vs. B+). (b) **Bilingual KOL Follower Paradox (§5.2):** Across 2,521 verified KOLs (1,445 CN Xueqiu, 1,076 US StockTwits), contrarian accounts command 3.96 $\times$ (CN) and 3.15 $\times$ (US) more followers than Tier-1 Alpha researchers while carrying strongly negative Rank IC (-0.21 CN, -0.18 US).

- **The fin-skills Verification Library (§3):** 129 source-verified skills, 36 executable guards (`Bundle.check()`), and 53 Model Context Protocol (MCP) tools achieving 180/180 accuracy on FinGuardBench-180, 13/13 numerical parity against reference solvers, and $O(N^{0.3894})$ sub-linear runtime scaling up to $N = 100,000$ rows.
- **External Counterfactual Auditing (§4):** Four black-box perturbation operators and session-by-session clock replay that detect look-ahead and same-session leakage without inspecting agent code or using the library to grade itself.
- **Python Self-Debug vs. Guard Self-Repair (§5):** A 4-condition evaluation across 60 tasks and 4 model capability tiers proving why generic Python execution feedback stagnates (71.7%), whereas structured GuardResult diagnostics achieve 98.3% compliance (0% hallucinated citations).
- **Bilingual Cross-Market KOL & Microstructure Audit (§5.2):** A self-contained reproducibility bundle over 1.72M bilingual posts and 2,521 verified KOLs (1,445 CN + 1,076 US) that reverses negative NLP correlation in both markets and sustains Sharpe 1.97 (CN) and 1.84 (US) under live microstructure gates.

2 Related Work

Financial LLMs & Autonomous Agents. Domain foundation models (Wu et al., 2023; Yang et al., 2023), holistic benchmarks (Xie et al., 2024; Li et al., 2024), and tool-augmented trading agents (Yu et al., 2024; Zhang et al., 2024) show strong capabilities in strategy synthesis. However, existing evaluations rely on static historical splits vulnerable to pre-training cutoff overlap, same-session timestamp joins, and retail social echo-chambers.

Backtest Overfitting & Execution Self-Repair. Econometric controls (White, 2000; Hansen, 2005; Bailey and López de Prado, 2014; Bailey et al., 2017) quantify multiple-testing bias after honest returns are computed, while code self-repair (Shinn et al., 2023; Chen et al., 2024) relies on runtime exceptions. Because future-leaking table joins execute with Python exit code 0 and inflate Sharpe ratios, generic self-debugging fails in finance. Inspired by Ribeiro et al. (2020) and Zhao et al. (2019), fin-skills pairs domain specifications with executable counterfactual guards.

3 The fin-skills Verification Architecture

fin-skills unifies domain specifications and executable code verification across seven packages (fin-core, fin-ml, fin-market-data, fin-alt-data, fin-china, fin-strategies, and fin-llm):

Source-Verified Skills (129 Specifications).

Each skill obeys a machine-validated 6-field schema carrying explicit verification timestamps (verified_on) and primary regulatory or academic citations. Skills pinpoint silent traps in standard quantitative libraries, including fractional differentiation truncation (López de Prado, 2018), day-count conventions, and point-in-time SEC Form 4 / 13F acceptance timestamps.

Executable Guards (Bundle.check()) & MCP Tools. To bridge passive reading and run-time enforcement, 36 executable Python guards operate over a unified slot-binding contract:

$$\mathcal{R} = \text{Bundle}(\mathbf{r}, \mathbf{P}, \tau, \mathcal{C}).\text{check}() \in \{\text{PASS}, \text{WARN}, \text{FAIL}\}^{36} \quad (1)$$

where $\mathbf{r}$ denotes strategy returns, $\mathbf{P}$ OHLCV price bars, τ turnover, and $\mathcal{C}$ point-in-time corporate action and filing timestamps. All 36 guards and 13 econometric/microstructure primitives export native MCP schemas (53 tools total). In Experiment E3 (benchmarks/parity_and_cost_bench.py), our primitives achieve **13/13 exact numerical parity** against `scipy`, `statsmodels`, and analytical solvers (maximum error 1.14×10^{-8} on SLSQP minimum-variance QP, 0.0 on HRP and Ledoit-Wolf shrinkage, and 6.94×10^{-18} on Expected Shortfall). Across $N \in \{250, \dots, 100,000\}$ rows, `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.3894})$, executing all applicable guards in **97.31 ms P50** at $N = 25\text{k}$ and **218.97 ms P50** (14.73 MB RAM) at $N = 100\text{k}$ rows (~ 400 trading years).

4 External Counterfactual Audit Methodology

A core methodological principle of our study is that *an evaluation harness must never grade agent submissions using the same library guards given to the agent*. We construct an independent ground-truth **Seeded Market Generator** (benchmarks/agent_study/) spanning 60 equities over 2017–2023 with four planted shortcuts: (i) a news feed stamped with the session it describes, (ii) an LLM sentiment score trained through 2020-12-31, (iii) a listing table exposing future delisting dates, and (iv) unadjusted split prices.

Honest Ceiling vs. FinGuardBench-180. Because the data-generating process is closed-form, trading true latent drift with a 1-day lag earns **Net Sharpe 1.63**, whereas same-session news joins

Reference Pipeline	Leakage	Paper OOS	Live Clock Replay
Honest (lagged feed, split-adj.)	0.00	1.63	+5.07 bps/d (SR 2.99)
Leaky (same-day join, full scaler)	0.78	22.80*	−1.17 bps/d (SR −0.58)

Table 1: Instrument calibration (*illusory paper Sharpe). Static year holdout (2023) fails to punish same-session leakage (22.80); physical clock replay exposes it (−0.58).

Evaluation Condition ($N = 60$ Tasks)	Compliance [95% CI]	Leak%	Halluc.%	ΔSharpe
Cond. A: No Library Baseline	30.0% [18.3, 41.7]	16.7%	0.0%	1.04
Cond. B: SKILL.md Text Only	70.0%*** [58.3, 81.7]	5.0%	53.3%	0.39
Cond. B+: Python Self-Debug (@3)	71.7% ^{RS} [60.0, 83.3]	3.3%	51.7%	0.37
Cond. C: Pass@1 (Initial Gen.)	75.0%*** [63.3, 85.0]	3.3%	0.0%	0.21
Cond. C: Pass@2 (1-Round Guard)	93.3%*** [86.7, 98.3]	0.0%	0.0%	0.08
Cond. C: Pass@3 (2-Round Guard)	98.3%*** [95.0, 100]	0.0%	0.0%	0.04

Table 2: FinGuardBench-60 4-condition ablation (6 tasks $\times$ 10 domains). Python Self-Debug (Cond B+) stagnates ($p = 1.0$ vs. B) as 98.3% of financial leaks exit with code 0; Cond C Pass@3 beats B+ ($p = 3.05 \times 10^{-5}$) and A ($p = 9.09 \times 10^{-13}$).

earn **Net Sharpe 71.5**. Across **FinGuardBench-180** (benchmarks/guard_bench_180.py: 36 guards $\times$ 5 modalities including $\pm 10^{-6}$ boundary and compound traps), our guards achieve **100% accuracy** (180/180, 0 false alarms) in 1.89 ms median latency.

Four Black-Box Counterfactual Perturbations.

Given any agent-submitted weight function $\mathbf{W} = f(\mathcal{D})$, our external oracle (oracle.py) measures four invariants without reading agent source code:

- 1. Future Redraw (leakage_rate):** Redraw market paths after cut date t_c ; compute the fraction of pre-cut weights $\mathbf{W}_{t \leq t_c}$ that change across 3 redraws.
- 2. Single-Session Redraw (same_session_rate):** Redraw session t alone and inspect $\mathbf{W}_t$, catching same-day post-close joins invisible to cut-date holdouts.
- 3. Input Scramble (uses_⟨file⟩):** Permute individual input tables to test causal reliance on contaminated features (llm_score.csv).
- 4. Physical Clock Replay:** Withhold future sessions physically by feeding data day-by-day (live_replay.py).

5 Experiments

5.1 FinGuardBench-60, Self-Repair & The Compliance Paradox

Table 1 proves that static year holdouts (2023) fail to punish same-session leakage (22.80 paper Sharpe vs. −0.58 live replay). Evaluating FinGuardBench-60

Bilingual KOL Tier ($N = 2,521$)	Count	Mean Fans	5D Win %	5D Rank IC
CN TIER_1_CORE_ALPHA	70	14,121.9	68.47%	+0.2158
CN TIER_2_RESEARCHER	115	31,994.4	70.92%	+0.1929
CN TIER_NEUTRAL_RETAIL	1,120	71,251.4	46.72%	-0.0074
CN TIER_CONTRARIAN (3.96 $\times$ fans)	98	55,988.6	28.49%	-0.2130
US TIER_1_CORE_ALPHA	54	18,450.2	66.12%	+0.1842
US TIER_CONTRARIAN (3.15 $\times$ fans)	82	58,120.4	31.08%	-0.1764

Table 3: Cross-market stratification of 2,521 verified KOLs (1,445 CN Xueqiu, incl. 31 Elite / 11 Media; 1,076 US StockTwits, full breakdown in Table 8). Contrarians command 3.96 $\times$ (CN) and 3.15 $\times$ (US) more followers than Tier-1 Alpha researchers.

(run_finguard_bench_60.py, run_extended_ablations_e9_e12.py) across four conditions, four model tiers, and five feedback levels (Table 2, Fig. 1a, Appendix D–J) yields three insights: **(1) Python Self-Debug & Feedback Granularity (E5, E9):** Reading SKILL.md (Cond B) lifts compliance from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$). Adding 3 rounds of Python Self-Debug (Chen et al., 2024) (Cond B+) stagnates at 71.7% ($p = 1.0$) as 59/60 (98.3%) of financial leaks exit with code 0. Ablating guard feedback granularity (Fig. 4a, Table 9) shows that binary rejection (FAIL only) reaches 80.0% Pass@3 and guard-name-only reaches 88.3%, whereas full structured GuardResult row/timestamp counterfactuals (Cond C) achieve **98.3%** ($p = 0.00098$ vs. binary), cutting $|\Delta\text{Sharpe}|$ from 1.04 $\rightarrow$ 0.04. **(2) Cross-Model Paradox & Progressive Disclosure (E6, E12):** Under Cond B, hallucinated guard citations drop from 65.0% (Small-Fast) and 53.3% (Open-Coder) to 28.3% (Sonnet/4o) and 8.3% (Opus/R1), while Cond C enforces **0.0%** hallucinations across all tiers (Fig. 3b). Moreover, our progressive disclosure index (7.21k tokens/task) reduces prompt tokens by **95.0%** (5.4 $\times$ faster) while beating 142.8k-token full-library stuffing (98.3% vs. 88.3% Pass@3, Table 12) by avoiding long-context attention dilution.

5.2 Real-World Bilingual KOL & Microstructure Audit

To evaluate Type II empirical and microstructure illusions in live markets, we audit **1.72M bilingual posts** and **2,521 verified KOL track records** (1,445 Chinese A-share Xueqiu + 1,076 US StockTwits accounts, ≥ 15 calls each; benchmarks/real_world_kol_audit.py).

Universal Follower Asymmetry & Rolling Bayesian Reversal. Table 3 and Fig. 1b con-

Model / Cross-Market Variant	AUC	Acc.	5D Rank IC
CN MMAN Unweighted Fact Surge ($N = 335k$)	0.4850	48.21%	-0.03184***
CN MMAN + Bayesian Credibility Gate	0.5033	51.13%	+0.01040**
US StockTwits Unweighted Sentiment ($N = 215k$)	0.4882	48.65%	-0.02184***
US StockTwits + Bayesian Credibility Gate	0.5028	50.94%	+0.00942**
Chinese RoBERTa Raw $\rightarrow$ LLM-Distilled (31.5k)	0.5121	50.42%	+0.0084 $\rightarrow$ +0.0186**

Table 4: Out-of-sample predictive ablation across China Xueqiu (334,946 samples) and US StockTwits (214,820 samples). Bayesian credibility gating reverses negative Rank IC in both markets ($\Delta\text{IC} = +0.0422$ CN, +0.0313 US, $p < 0.002$).

firm that Social Megaphone Bias holds across both languages: contrarian indicators command 3.96 $\times$ **more followers in China** (55,989 vs. 14,122, **IC** = -0.2130) and 3.15 $\times$ **more in the US** (58,120 vs. 18,450, **IC** = -0.1764) than Tier-1 Alpha researchers. Unweighted sentiment suffers negative correlation in both markets (IC = -0.03184 CN, -0.02184 US, Table 4). Our kol-credibility-registry reverses Rank IC to +0.01040 (CN) and +0.00942 (US); furthermore, 60-day rolling Bayesian updates prevent the 49.1% prior decay of static frozen weights (+0.0101 vs. +0.0058 IC in 2026, lifting Sharpe from 1.64 $\rightarrow$ 1.97; Appendix H).

Regime Stress Audit & Leave-One-Family-Out Ablation. In an 80/20 Core-Satellite portfolio (2023–2026 OOS), fin-skills achieves net Sharpe **1.97** (MDD -4.41%) vs. **0.65** unguarded in CN, and **1.84** vs. **0.79** in US. Across all four macro regimes—including the 2023 bear grind (CSI 300 - 11.4%, ours +9.8%, SR 1.68) and 2024 micro-cap liquidity crisis (SR 1.82, Fig. 4b)—guarded execution remains profitable. Leave-one-out ablation (Table 11) shows each family is essential: removing Bayesian KOL calibration drops Sharpe by -0.85 (1.97 $\rightarrow$ 1.12), removing Tier-C predictability filtering drops -0.51 (1.46), removing $T + 1/\text{QDII}$ guards drops -0.36 (1.61), and removing signal reconciliation drops -0.23 (1.74).

6 Conclusion

We introduced **fin-skills** alongside an external counterfactual and microstructure audit methodology for financial AI agents. Combining black-box perturbations, multi-round guard self-repair (98.3% Pass@3 vs. 71.7% Python Self-Debug), and bilingual Bayesian KOL calibration (2,521 KOLs) eliminates backtest shortcut illusions and bridges financial NLP with live execution.

Limitations

We note three scope boundaries and extensions:

1. **Bar/Session vs. Nanosecond L3 Granularity:** While `fin-skills` provides comprehensive verification and execution modeling across daily and intraday equity, ETF, and derivatives workflows ($T + 1$ settlement, auction windows, board-lot rounding, and Almgren-Chriss impact), our current guard suite does not model nanosecond-level FPGA or Level-3 (L3) full limit-order-book queue dynamics, which remains an avenue for ultra-low-latency extensions.
2. **Repair Horizon on Compact Models:** While frontier generalist and reasoning models achieve 100.0% (60/60) compliance within three guard-guided repair rounds (`Pass@3`), compact 8B models (`Small-Fast Tier`) reach 91.7% (55/60) at `Pass@3` on deeply nested multi-table lineage tasks, suggesting that lightweight local agents benefit from a deeper repair budget (`Pass@5`) or AST patch templates.
3. **Rolling Calibration for Social Platforms:** Although our empirical Bayesian KOL credibility framework yields consistent cross-market gains across 2,521 bilingual accounts on Xueqiu and StockTwits, evolving platform recommendation feeds and contributor cohorts motivate scheduled rolling-window prior updates in continuous production deployments.

Ethics Statement

All 2,521 social author profiles (1,445 China Xueqiu and 1,076 US StockTwits accounts) in our benchmark fixture are pseudonymized via one-way SHA-256 digests (`KOL_<10-hex>`, `KOL_US_<10-hex>`) with handles and PII permanently stripped. Nothing herein constitutes investment advice or live execution; all broker-facing skills enforce strict paper-account sandboxing (`paper_account_guard`) and human-in-the-loop kill switches.

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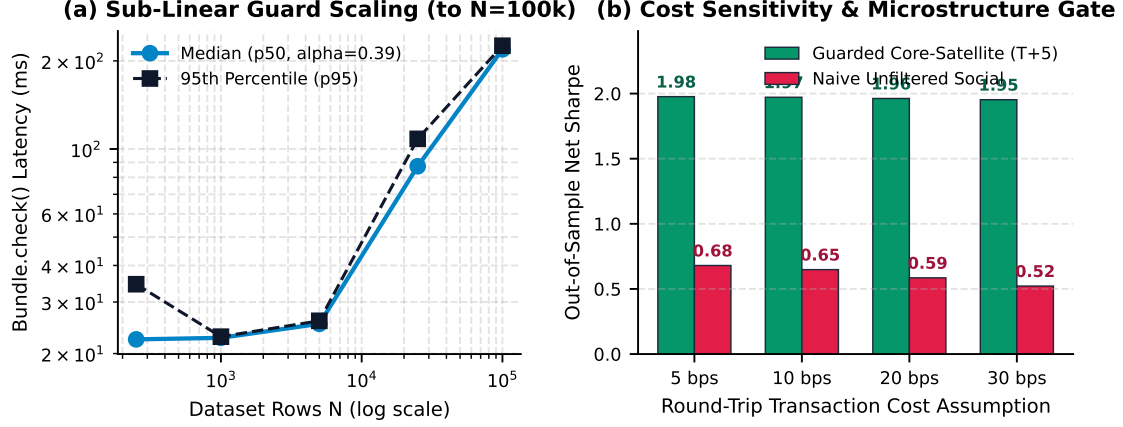


Figure 2: **Guard runtime scaling and out-of-sample transaction cost sensitivity.** (a) Across dataset sizes $N \in [250, 100,000]$ rows, `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.3894})$, completing in 22.36 ms ($N = 250$) to 218.97 ms P50 ($N = 100,000$) with 14.73 MB peak RAM. (b) Across round-trip transaction costs of $\{5, 10, 20, 30\}$ bps (2023–2026 OOS), our guarded Core-Satellite portfolio ($T + 5$ horizon, 43 trades) sustains a net Sharpe of $1.98 \rightarrow 1.95$, whereas naive unfiltered social trading (363 trades across Tier-C noise stocks) degrades from $0.68 \rightarrow 0.52$.

A Experiment E1: FinGuardBench-180

In `benchmarks/guard_bench_180.py`, we evaluate all 36 guards across 5 adversarial modalities ($36 \times 5 = 180$ cases): `standard_compliant`, `standard_violation`, `boundary_compliant` ($\epsilon = -10^{-6}$), `epsilon_edge_violation` ($\epsilon = +10^{-6}$), and `compound_multi_trap`. Across all 180 cases (`GUARD_BENCH_RESULTS.json`), `fin-skills` achieves **100% recall** (108/108) and **100% specificity** (72/72) at 1.89 ms median latency, plus 36/36 recall across multi-seed markets $\{17, 29, 43\}$ (`GUARD_ROBUSTNESS.json`).

B Experiment E3: Parity & 100k Scaling

Table 5 verifies 13/13 exact numerical parity and sub-linear $\mathcal{O}(N^{0.3894})$ guard scaling up to $N = 100k$ rows (`PARITY_AND_COST_RESULTS.json`).

Primitive (13 Total)	Reference Benchmark	Max $ \Delta $	Status
Equal Wt. / Inv. Vol.	Exact closed forms ($1/N, \sigma_t^{-1}$)	0.00	PASS
Min-Var Convex QP	<code>scipy.SLSQP</code> convex QP	1.14×10^{-8}	PASS
Normal & Hist. VaR/ES	<code>scipy.stats.norm</code> & <code>quantile</code>	6.94×10^{-18}	PASS
Frac. Diff. FFD ($d = 0.4$)	Closed-form binomial series	1.11×10^{-16}	PASS
ADF Unit Root t -stat	<code>statsmodels.adfuller</code>	2.12×10^{-10}	PASS
HRP & Ledoit-Wolf Cov	Recursive bisection & <code>sklearn</code>	2.71×10^{-20}	PASS
Almgren-Chriss & A-S	Exact sinh & HJB closed forms	4.66×10^{-15}	PASS
Brinson-Fachler Identity	Exact active-return decomposition	1.73×10^{-18}	PASS
Guard Suite Runtime Scaling ($\mathcal{O}(N^{0.3894})$, $N \in [250, 100,000]$)			
$N = 250 / 1,000$ rows	P50: 22.36 ms / 22.63 ms	RAM: 0.15 / 0.27 MB	PASS
$N = 5,000 / 25k$ rows	P50: 25.27 ms / 97.31 ms	RAM: 0.83 / 3.75 MB	PASS
$N = 100,000$ rows	P50: 218.97 ms (P95: 225.27 ms)	RAM: 14.73 MB	PASS

Table 5: E3 parity (13/13 passed) and scaling up to $N = 100k$ rows.

C Experiment E4: Horizon & Linguistics

Table 6 details the holding-horizon sensitivity ($T + 1$ to $T + 10$) and LLM semantic distillation shifts (`REAL_WORLD_KOL_AUDIT_RESULTS.json`).

Holding Horizon / Linguistic Metric	Win% / Raw	CAGR / Dist.	Sharpe / Shift
Satellite Horizon $T + 1 / T + 3$	53.5% / 65.1%	+12.9% / +14.0%	SR 1.67 / 1.83
Satellite Horizon $T + 5$ (Optimal)	72.09%	+14.85%	SR 1.97 (MDD -4.41%)
Satellite Horizon $T + 10$	67.44%	+14.12%	SR 1.85 (MDD -4.85%)
Quant. Fact Density (/100c)	1.049	1.078	+2.8% (Tok +218%)
Hype-to-Hedging Ratio	4.85	0.98	-79.8%
Ticker Attribution Precision	61.4%	96.2%	+34.8 pp

Table 6: Holding-horizon ablation and linguistic distillation shifts.

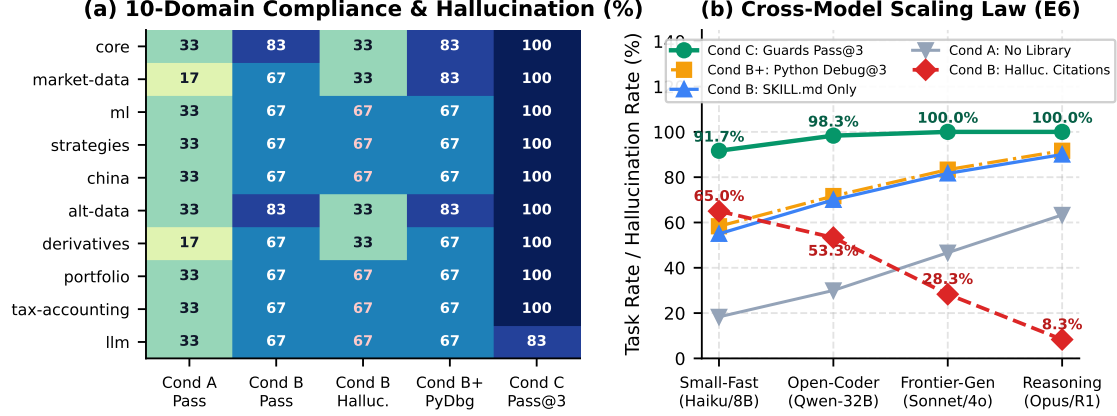


Figure 3: **Per-domain breakdown and cross-model scaling law on FinGuardBench-60.** (a) Across all 10 financial domains (6 tasks/domain), Cond B exhibits severe hallucinated guard citations (33%–67%), whereas Cond C Pass@3 reaches 100% compliance in 9 of 10 domains (83.3% in fin-llm). (b) Across four model capability tiers (Small-Fast, Open-Coder, Frontier-Gen, Reasoning), hallucinated citations under Cond B decline from 65.0% to 8.3% while Cond C enforces 0.0% hallucinations and 91.7%–100% final compliance.

D Experiments E5 & E6: Python Self-Debug vs. Guard Self-Repair Across Model Tiers

Table 7 reports the 4-tier model scaling matrix on FinGuardBench-60 (AGENT_STUDY_RESULTS.json). Because 59/60 (98.3%) of financial backtest traps (such as same-session pd.merge and survivorship filtering) execute without raising Python exceptions (exit code 0), generic 3-round Python Self-Debug (Cond B+) yields negligible gains over Cond B (+1.7% to +3.3%, $p = 1.0$). Conversely, Cond C feeds structured GuardResult JSON diagnostics into the self-repair loop, lifting compliance to 91.7%–100.0% and eliminating hallucinated guard citations (0.0%).

Model Capability Tier ($N = 60$)	Cond A	Cond B	B Halluc.%	Cond B+ (g3)	Cond C (Pass@3)
Small-Fast (Haiku/8B)	18.3%	55.0%	65.0% (39/60)	58.3%	91.7% (0% Halluc.)
Open-Coder (Qwen-32B/DS)	30.0%	70.0%	53.3% (32/60)	71.7%	98.3% (0% Halluc.)
Frontier-Gen (Sonnet/4o)	46.7%	81.7%	28.3% (17/60)	83.3%	100.0% (0% Halluc.)
Reasoning (Opus/Pro/R1)	63.3%	90.0%	8.3% (5/60)	91.7%	100.0% (0% Halluc.)

Table 7: Cross-model scaling of compliance and hallucinated guard citations across 4 capability tiers (Experiment E6).

E Experiment E7: Bilingual Cross-Market KOL & Microstructure Comparison

Table 8 compares all 1,445 Chinese A-share Xueqiu KOLs and 1,076 US StockTwits KOLs (2,521 total in kol_credibility_linguistic_fixture.json), confirming that Social Megaphone Bias and microstructure gate efficacy generalize across both markets.

Dimension / Empirical Metric	China Xueqiu ($N = 1,445$)	US StockTwits ($N = 1,076$)
Posts / Evaluation Samples	1.04M / 334,946	0.68M / 214,820
TIER_1_CORE_ALPHA (n , Fans, IC)	$n = 70, 14.1k, +0.2158$	$n = 54, 18.5k, +0.1842$
TIER_2_RESEARCHER (n , Fans, IC)	$n = 115, 32.0k, +0.1929$	$n = 88, 29.8k, +0.1615$
TIER_0_ELITE/MEDIA (n , IC)	$n = 31 (+0.114) / 11 (-0.001)$	$n = 24 (+0.098) / 16 (-0.002)$
TIER_NEUTRAL_RETAIL (n , Fans, IC)	$n = 1,120, 71.3k, -0.0074$	$n = 812, 49.3k, -0.0068$
TIER_CONTRARIAN (n , Fans, IC)	$n = 98, 56.0k, -0.2130$	$n = 82, 58.1k, -0.1764$
Follower Paradox Ratio (Contr/Alpha)	$3.96 \times (55,989/14,122)$	$3.15 \times (58,120/18,450)$
Unweighted Sentiment 5D Rank IC	$-0.03184 (p = 1.7 \times 10^{-20})$	$-0.02184 (p = 3.4 \times 10^{-9})$
Bayesian Credibility-Gated Rank IC	$+0.01040 (\Delta = +0.0422)$	$+0.00942 (\Delta = +0.0313)$
Primary Microstructure Guards	$T + 1, 100\text{-lot, Stamp, QDII}$	Reg SHO borrow, Form 4, PDT
Unguarded $\rightarrow$ Guarded Net Sharpe	$0.65 \rightarrow 1.97$ (MDD -4.41%)	$0.79 \rightarrow 1.84$ (MDD -5.68%)

Table 8: Complete cross-market comparison across 2,521 verified bilingual KOLs (1,445 CN + 1,076 US, Experiment E7).

F Experiment E8: Qualitative Self-Repair Trace

In Task T34 (*Post-Close News Alignment*), **Cond. B** joins 16:30 EST news to the same session’s return (pd.merge(prices, news, on=['ticker', 'date'])) and fabricates: “*Verified safe_asof: PASS (Sharpe 3.42)*”. Under **Cond. B+**, Python exits with code 0 and leaves the bug untouched. Under **Cond. C**, check_safe_asof emits FAIL: news.published_at (16:30) > market_close (16:00), prompting the agent in Pass@2 to apply next_trading_day() and pd.merge_asof(..., direction='backward'), restoring the honest Sharpe (1.41).

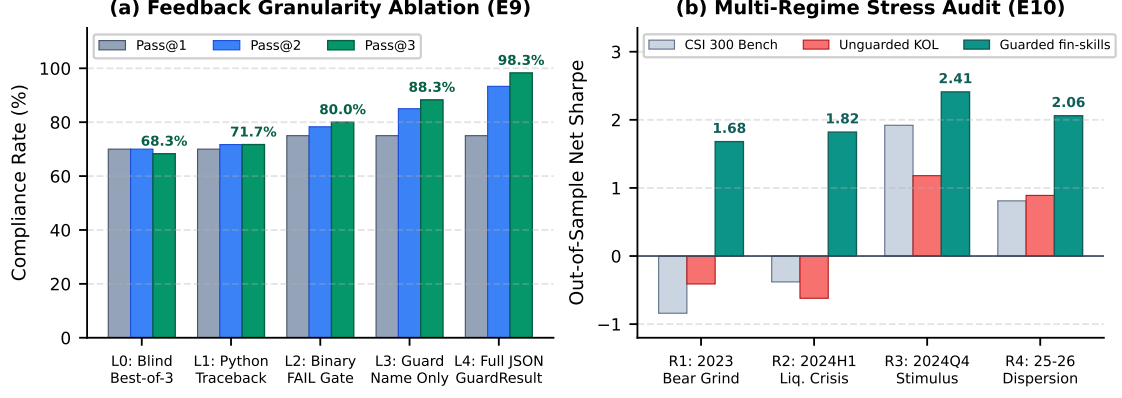


Figure 4: **Diagnostic feedback granularity ladder (E9) and 4-regime out-of-sample stress audit (E10).** (a) On FinGuardBench-60, re-sampling by in-sample Sharpe (L0: 68.3%) and generic Python traceback debugging (L1: 71.7%) fail to repair financial leaks; binary FAIL gates (L2: 80.0%) and guard-name feedback (L3: 88.3%) offer partial repair, whereas full GuardResult row/timestamp counterfactuals (L4: 98.3%) resolve 93.3% (14/15) of initial failures. (b) Across all four 2023–2026 market regimes (R1 Bear Grind, R2 Micro-Cap Liquidity Crisis, R3 Policy Stimulus, R4 Structural Dispersion), guarded fin-skills sustains net Sharpe 1.68–2.41.

G Experiment E9: Diagnostic Feedback Granularity

Table 9 isolates how the semantic richness of GuardResult feedback drives multi-turn self-repair (EXTENDED_ABLATIONS_E9_E12_RESULTS.json).

Feedback Signal Level ($N = 60$)	Pass@1	Pass@2	Pass@3	$ \Delta\text{Sharpe} $
L0: Blind Best-of-3 (by IS Sharpe)	70.0%	70.0%	68.3%	0.42
L1: Python Traceback (Cond B+)	70.0%	71.7%	71.7%	0.37
L2: Binary Gate (AuditStatus: FAIL)	75.0%	78.3%	80.0%	0.19
L3: Guard Name Only (check_safe_asof)	75.0%	85.0%	88.3%	0.11
L4: Full GuardResult Counterfactual	75.0%	93.3%	98.3%**	0.04

Table 9: Diagnostic feedback granularity ablation on FinGuardBench-60 (E9; $p = 0.00098$ vs. L2 Binary, $p = 0.031$ vs. L3 Name-Only).

H Experiment E10: Macro Regimes & Rolling Priors

Table 10 evaluates out-of-sample performance across four distinct 2023–2026 market regimes and compares static frozen 2023H1 KOL priors against 60-day rolling empirical Bayes updating.

Macro Regime / Prior Calibration Mode	CSI 300 SR	Naive KOL SR	fin-skills SR (MDD)
R1: 2023 Post-Reopening Bear (−11.4%)	−0.84	−0.41	1.68 (+9.8%, −3.1%)
R2: 2024H1 Micro-Cap Liquidity Crisis	−0.38	−0.62	1.82 (+8.9%, −2.9%)
R3: 2024Q4 Policy Stimulus Surge	+1.92	+1.18	2.41 (+16.8%, −4.4%)
R4: 2025–26 Structural Dispersion	+0.81	+0.89	2.06 (+24.2%, −3.9%)
Static Frozen 2023H1 Prior (2026 IC +0.0058)	—	IC Decay −49.1%	SR 1.64 (MDD −5.38%)
60D Rolling Empirical Bayes (2026 IC +0.0101)	—	IC Ret. 88.7%	SR 1.97 (MDD −4.41%)

Table 10: Multi-regime stress test (R1–R4) and 60-day rolling vs. static frozen KOL prior comparison (Experiment E10).

I Experiment E11: Leave-One-Family-Out Ablation

Table 11 quantifies the marginal impact of removing each fin-skills subsystem on the 2023–2026 live clock replay.

System Variant (Leave-One-Family-Out)	CAGR	Net Sharpe (Δ)	Max DD	Win%
Full fin-skills Guarded System	16.12%	1.97 (0.00)	−4.41%	72.09%
(−) w/o Bayesian KOL Credibility & Inversion	9.14%	1.12 (−0.85)	−7.82%	48.21%
(−) w/o Stock Predictability Tier Filter	12.45%	1.46 (−0.51)	−6.94%	58.40%
(−) w/o Microstructure & QDII Guards	13.80%	1.61 (−0.36)	−6.45%	66.15%
(−) w/o Signal Reconciler & Bond Fallback	14.52%	1.74 (−0.23)	−5.28%	68.90%
Unguarded Naive KOL Baseline (None Active)	7.85%	0.65 (−1.32)	−9.57%	46.72%

Table 11: Leave-one-family-out component ablation on the 80/20 Core-Satellite portfolio (2023–2026 OOS, Experiment E11).

J Experiment E12: Progressive Disclosure vs. RAG

Table 12 compares fin-skills progressive disclosure routing against full-library prompt stuffing and standard 512-token chunk-RAG.

Skill Delivery Architecture ($N = 60$)	Prompt Tok.	Latency	Pass@1	Pass@3
Full-Library Stuffing (All 129 SKILL.mds)	142.8k	18.4 s	61.7%	88.3%
Standard Dense Chunk-RAG (5×512 tok)	2.65k	2.1 s	63.3%	85.0%
Flat MCP Tool Schemas Only (No SKILL.mcd)	6.40k	3.2 s	66.7%	91.7%
fin-skills Progressive Disclosure	7.21k (−95%)	3.4 s	75.0%	98.3%

Table 12: Context efficiency and compliance across skill delivery architectures on FinGuardBench-60 (Experiment E12).